

Project Proposal

Question Answering System using RAG

Team Information

Name: Humna Imran

Registration No: 2023-BS-AI-017

Email: 2023-bs-ai-017@tuf.edu.pk

Name: Muhammad Wahaj

Registration No: 2023-BS-AI-007

Email: 2023-bs-ai-007@tuf.edu.pk

Problem Statement

A Document-Based Question Answering System using RAG

University students, staff, and faculty often spend excessive time searching through lengthy policy documents, course catalogs, or rulebooks to find simple answers. Traditional keyword searches are often ineffective at understanding the context of a user's question, leading to frustration and wasted time.



Motivation & Industry Relevance

This project creates a practical, time-saving tool that benefits the university ecosystem (students and administration). By allowing users to get instant, accurate answers directly from official documents, we can improve the student experience and reduce the administrative burden of repeatedly answering standard queries.

Literature Review (Brief)

Keyword-based search

Traditional keyword-based search systems struggle with understanding the semantic context of user queries



Large Language Models

Standard Large Language Models (LLMs) can hallucinate or lack access to specific, private institutional data



Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) solves this by fetching relevant document chunks before generating an answer. Our approach will implement a lightweight RAG system that improves upon standard search by providing direct, context-aware answers.

Proposed Methodology

- Type of problem: NLP / Question Answering System.
- Pipeline: We will build a RAG pipeline that extracts text from PDFs, chunks it, creates embeddings, and uses a generative model to synthesize answers based on the retrieved context.
- Model Comparison: To satisfy project requirements, we will compare two different open-source embedding models (e.g., all-MiniLM-L6-v2 vs. bge-small-en) to evaluate which model retrieves the most accurate context for our documents

Dataset Description

- Source and Size: A custom, small-scale dataset consisting of 5-10 official University Handbooks, Prospectuses, and Policy PDFs.
- Data preprocessing: Extracting text from PDFs, cleaning formatting, and splitting the text into manageable chunks (e.g., 500-word segments) for the vector database.

Tools & Technologies

- Programming language: Python.
- Deep learning framework: PyTorch and Hugging Face for accessing open-source models.
- Deployment tools: Streamlit to build an interactive web interface.

Evaluation Plan

Evaluate the system's retrieval

- We will evaluate the system's retrieval performance by comparing the accuracy of the two embedding models.
- We will use metrics such as F1 Score (for answer similarity) and human evaluation to measure the relevance and clarity of the generated responses.

Expected Outcomes

- By the end of the project, we expect to deliver a functional RAG-based AI assistant.
- Deliverables will include a working Streamlit prototype where a user can upload a PDF and ask questions, trained/configured model files, and a performance comparison report of the two models.

Project Timeline

Proposal submission

Week 1–4

Proposal submission, problem statement definition, collecting dataset (PDFs), and initial text preprocessing.

Building the basic RAG pipeline

Week 5–8

Building the basic RAG pipeline, implementing the first baseline model, and presenting initial results for the Mid Evaluation

Second model for comparison

Week 9–12

Implementing the second model for comparison, conducting evaluations using appropriate metrics, and fine-tuning

Streamlit deployment

Week 13–16

Developing the Streamlit deployment demo, recording the 3-5 minute demo video, and preparing the IEEE format final report and presentation slides.

Division of Responsibilities

- Member 1: Responsible for dataset preparation, text chunking, implementing the embedding models, and evaluating retrieval metrics.
- Member 2: Responsible for generative model integration, developing the Streamlit deployment prototype, and formatting the final report.

Supervisor Approval

Signature: _____

Date: _____

Thank You

Submitted to Dr. Guffran